

CMR Specialist Execution Phase

From: CMR Specialist Product Architecture Design Specification, Execution draft, 31 August 2026.

Purpose	Build vertical slices in dependency order. Every phase ends in a user-visible capability and a testable operational outcome.	Immediate focus	Create durable case and knowledge systems first, then move AI flows inside those auditable boundaries.
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Implementation Plan

PHASE	BUILD	EXIT CHECK
0 Foundation	Repository layout, environments, container images, CI, secrets, health checks.	API and worker deploy from one commit.
1 Durable core	PostgreSQL schema, migrations, users, cases, events, audit log, role access.	Case survives restart and has a complete timeline.
2 Customer side	Sign-in, chat, case creation, case history, attachments, feedback.	A customer can ask, continue, and track a case.
3 Support console	Queues, filters, assignment, notes, state changes, customer history.	Staff can resolve a case without database access.
4 Knowledge	Upload, version, parse/OCR, chunk, embed, validation, publish, rollback.	A new PDF can be published and cited safely.
5 AI + traceability	Binary routing, grounded generation, retrieval trace, model run record, fallback.	Every answer can be reconstructed from stored metadata.
6 Operations	Metrics, logs, traces, dashboards, alerts, backup checks, runbooks.	An operator can detect and diagnose failure.
7 Launch	Security review, load test, recovery drill, staged rollout, support training.	SLOs pass at 2x expected peak traffic.

Suggested Team And Timing

One strong Python engineer	Approximately 6-8 weeks for a controlled pilot, provided identity, messaging and model integrations are straightforward.
Two engineers	Approximately 4-6 weeks when backend/platform and interface work progress in parallel with shared review.
Planning condition	These are delivery estimates, not commitments. External approvals, data migration and security review can extend the calendar.

Module Layout

app/api	Customer, support and operations routes	app/cases	Case commands, state and events
app/knowledge	Documents, versions and publishing	app/ai	Routing, retrieval, inference and validation
app/platform	Database, auth, audit and telemetry	worker/tasks	Ingestion, embeddings, notifications, exports
migrations	Versioned database changes	tests	Unit, integration, retrieval and load tests

Immediate Execution Order

1	Create the PostgreSQL schema and migrations for people, cases, case events, conversations, messages and audit events.
2	Move the existing binary router and grounded answer flow behind the new conversation and case application services.
3	Build the support case view and document publishing workflow before adding visual polish to the operations dashboard.
4	Deploy two API containers and one worker, instrument them, then load test the full answer path with realistic prompts and documents.

Execution note	Build the durable case and knowledge systems first. Let AI improve inside those boundaries without becoming the boundary.
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Source sections: 08 / Design Spec - Implementation Plan and 10 / Design Spec - Definition of Done, Immediate Execution Order.